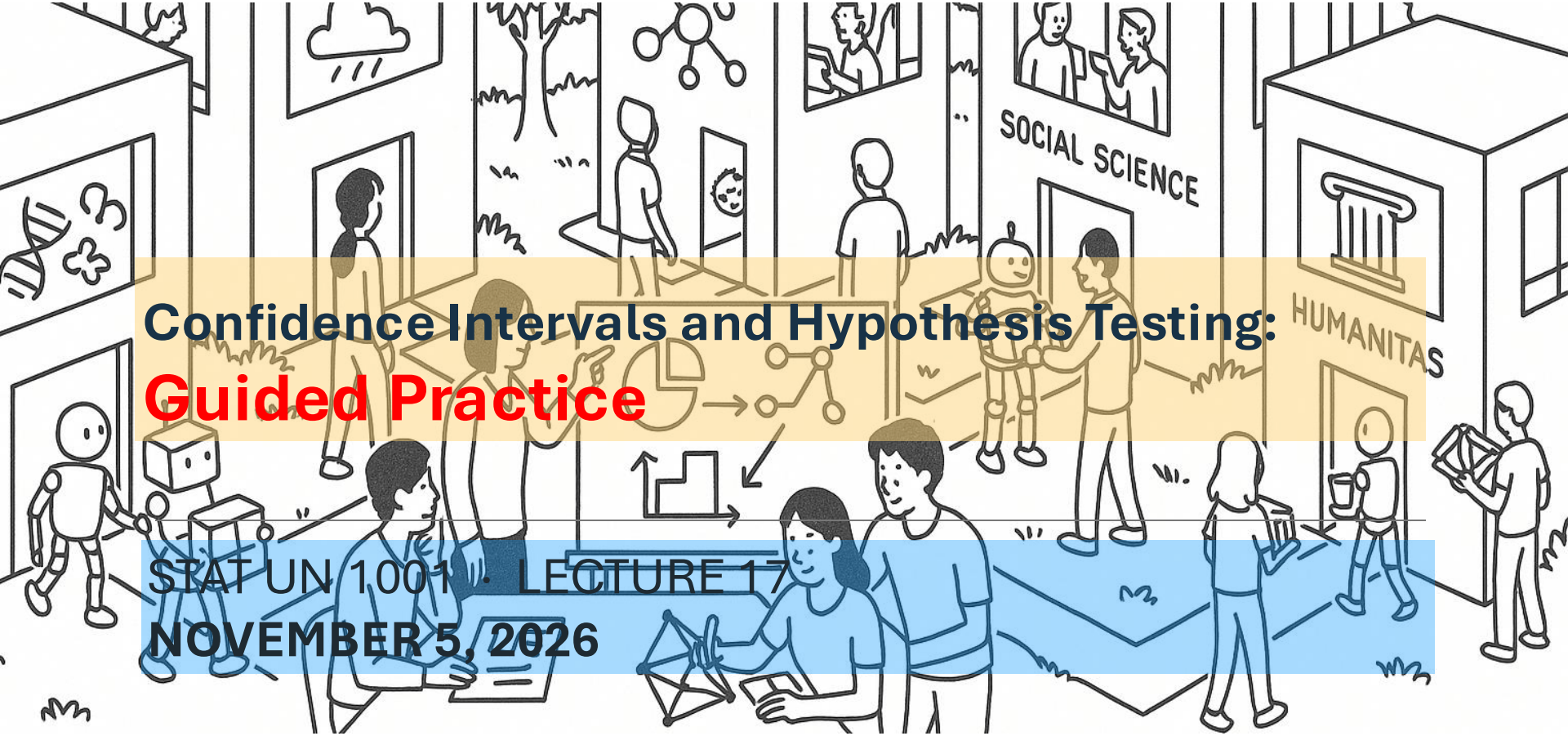


Confidence Intervals and Hypothesis Testing:

Guided Practice

STAT UN 1001 · LECTURE 17

NOVEMBER 5, 2026

Today's hands-on plan

Vocabulary reset	Which number does what?
Confidence intervals	Conditions, SE, margin, endpoints, interpretation
Hypothesis test	Hypotheses, test SE, z, P-value, decision
Connections	CI versus test; statistical versus practical importance
Exit ticket	One complete conclusion in context

Work in pairs. Write the setup before calculating; both partners should be ready to explain.

What your check-ins told us

45 anonymous Week 9 check-ins

Area requested	Students	Share
CI interpretation	13	29%
CI margin/endpoints	12	27%
P-value meaning	12	27%
H_o and H_a	10	22%
One- vs two-sided tests	41 wanted more practice	91%

The symbols

Symbol	Meaning	Used for
p	Unknown population proportion	The parameter we want to learn about
$\hat{p}$	Observed sample proportion	Point estimate; confidence-interval SE
p_o	Value asserted by H_o	Hypothesis Test's Sampling Distribution for $\hat{p}$: a normal dist. with mean and SE
z^*	Confidence multiplier	Margin of error
z	Standardized test statistic	Distance from p_o in null SE units
P-value	Probability under H_o	Evidence, not P(H_o is true)

Confidence interval workflow

1	**Conditions**	Random/representative; independence; at least 10 successes and 10 failures
2	**Estimate**	$\hat{p} = x / n$
3	**Standard error**	$SE = \sqrt{[\hat{p}(1 - \hat{p}) / n]}$
4	**Margin**	$ME = z^* \times SE$
5	**Interval**	$\hat{p} \pm ME$
6	**Interpret**	A plausible range for the population proportion p

Practice 1: a 95% confidence interval

A random sample of 250 students finds that 145 support extending the campus night-shuttle schedule. The university enrolls more than 2,500 students.

A. Check the conditions for a one-proportion confidence interval.

B. Compute $\hat{p}$ and its standard error.

C. Construct a 95% confidence interval.

D. Interpret the interval in context.

Use $z^* = 1.96$. Write a sentence—not just two endpoints.

Practice 1: worked answer

Conditions	Random sample; 250 is under 10% of enrollment; 145 successes and 105 failures are both at least 10.
Estimate	$\hat{p} = 145 / 250 = 0.580$
Standard error	$SE = \sqrt{[0.58(0.42) / 250]} = 0.0312$
Margin	$ME = 1.96(0.0312) = 0.0612$
Interval	**$0.580 \pm 0.0612 = (0.519, 0.641)$**

We are 95% confident that 51.9% to 64.1% of all students support the extension.

Practice 2: confidence versus precision

Use the same sample: n = 250, $\hat{p}$ = 0.580, SE = 0.0312.

Part A

Construct a 99% interval using $z^* = 2.576$.

Before computing: should it be wider or narrower than the 95% interval? Why?

Part B

A reported 95% interval is (0.46, 0.54).

Find its center, margin of error, and standard error.

A higher confidence level changes z^*, not the observed $\hat{p}$.

Practice 2: worked answer

Part A: 99% interval

ME = 2.576(0.0312) = 0.0804

0.580 ± 0.0804 = (0.500, 0.660)

It is wider because 99% confidence uses a larger multiplier.

Part B: work backward

Center = (0.46 + 0.54) / 2 = 0.50

ME = 0.54 − 0.50 = 0.04

SE = 0.04 / 1.96 = 0.0204

Confidence interval = estimate ± margin of error.

Hypothesis-test workflow

1	**Hypotheses**	What do H_0 and H_a claim?
2	**Conditions**	Does the one-proportion model fit under H_0?
3	**Null SE**	How much would $\hat{p}$ vary if p equaled p_0?
4	**Test statistic**	How many null standard errors is $\hat{p}$ from p_0?
5	**P-value**	How surprising are these data if H_0 is true?
6	**Conclusion**	Reject or fail to reject H_0; conclude in context.

One-sided and two-sided alternatives

Research question	Alternative	P-value tail(s)
Is p greater than 0.50?	H_a: $p > 0.50$	Right tail
Is p less than 0.50?	H_a: $p < 0.50$	Left tail
Is p different from 0.50?	H_a: p does not equal 0.50	Both tails

For a symmetric two-sided test, double the one-tail probability—not the standard error.

Practice 3A: set up a two-sided test

A random sample of 200 students finds that 116 support a proposed advising plan. Test whether the true support proportion differs from 0.50.

A. Define p and write H_0 and H_a.

B. Check the conditions under H_0.

C. Compute $\hat{p}$.

D. Compute the null standard error SE_0.

Which proportion belongs inside the test standard error: $\hat{p}$ or p_0?

Practice 3A: worked setup

Parameter	p = the true proportion of all students who support the plan.
Hypotheses	**H_0: p = 0.50 H_a: p does not equal 0.50**
Conditions	Random sample; population at least 2,000; under H_0, expected successes = 100 and failures = 100.
Estimate	$\hat{p}$ = 116 / 200 = 0.580
Null SE	**$SE_0 = \sqrt{[0.50(0.50) / 200]} = 0.0354$**

The test asks how $\hat{p}$ would behave in a world where H_0 is true.

Practice 3B: evidence and conclusion

Continue the advising-plan test. Use $\hat{p} = 0.580$ and $SE_0 = 0.0354$.

Given: $P(Z \geq 2.26) = 0.0119$.

A. Compute the test statistic z.

B. Find the two-sided P-value.

C. At $\alpha = 0.05$, reject or fail to reject H_0?

D. State the conclusion in context. What does the P-value not mean?

Practice 3B: worked answer

Test statistic	$z = (0.580 - 0.500) / 0.0354 = 2.26$
P-value	**$2(0.0119) = 0.0238$**
Decision	Because $0.0238 < 0.05$, reject H_0.
Conclusion	The sample gives statistically significant evidence that the true support proportion differs from 0.50.

The P-value is not the probability that H_0 is true.

It is a probability about possible data, calculated under H_0.

Confidence intervals and two-sided tests

For the same n = 200 and $\hat{p} = 0.580$:

95% confidence interval

$SE = \sqrt{[0.58(0.42)/200]} = 0.0349$

$0.580 \pm 1.96(0.0349)$

CI = (0.512, 0.648)

Two-sided test

H_0 value 0.50 is outside the 95% CI.

The two-sided test at $\alpha = 0.05$ rejects H_0.

The conclusions agree.

CI: plausible values for p.
Test: strength of evidence against one claimed value.

Practice 4: significance is not importance

A survey of 20,000 voters estimates support at $\hat{p} = 0.508$. Testing H_0: $p = 0.500$ gives P-value = 0.024.

A. Is the result statistically significant at $\alpha = 0.05$?

B. How large is the estimated difference from 0.500?

C. Is that difference necessarily important in practice?

D. What else would you want to know about the data collection?

A small P-value answers an evidence question —not a consequence question.

Practice 4: answer and decision errors

Statistically significant: $0.024 < 0.05$. Estimated difference: 0.008, or 0.8 percentage points.
Whether that matters depends on context and data quality.

New context:
H_0 says a safety message does not change helmet use;
H_a says it changes helmet use.

Type I error	Conclude the message changes helmet use when it actually does not.
Type II error	Fail to detect a real change in helmet use.

Name H_0 first. Then compare the decision with reality.

Exit ticket: one complete argument

Choose one of today's examples and complete these three sentences:

1. The population parameter is ...
2. The interval / P-value tells us ...
3. The conclusion we can support is ..., but we still do not know ...